
USER MANUAL

Urban Smart Parking Management System

CS 2011E — Database Management Systems
Winter Semester 2026 | Class CS02 | Team 1

Farzan Faisal · Elias Joby · Reja Fairooz · Amil K · Ronn Boban · Kona Bhuvaneswar Reddy
B2402179CS | B2402175CS | B240281CS | B240136CS | B240286CS | B240214CS

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1. Introduction

1.1 Purpose of This Manual

This User Manual provides step-by-step guidance for the Urban Smart Parking Management System (USPMS). It covers every feature available to both Administrators and Drivers — from account creation (for drivers), through billing and session history.

1.2 System Overview

USPMS is a web-based enterprise application that streamlines city parking. Drivers discover lots, reserve slots in real time, and track fees. Admins manage infrastructure, monitor sessions, and override billing where necessary.

⚡ Key Capabilities

- Real-time parking lot and slot availability
- Role-specific dashboards for Admins and Drivers
- Automated billing triggered on slot release
- JWT-secured authentication gateway
- Atomic MySQL transactions prevent double-booking

1.3 Intended Audience

- ▶ Administrators managing parking lots and user accounts
- ▶ Drivers reserving and managing parking slots
- ▶ System operators performing maintenance or testing

1.4 System Requirements

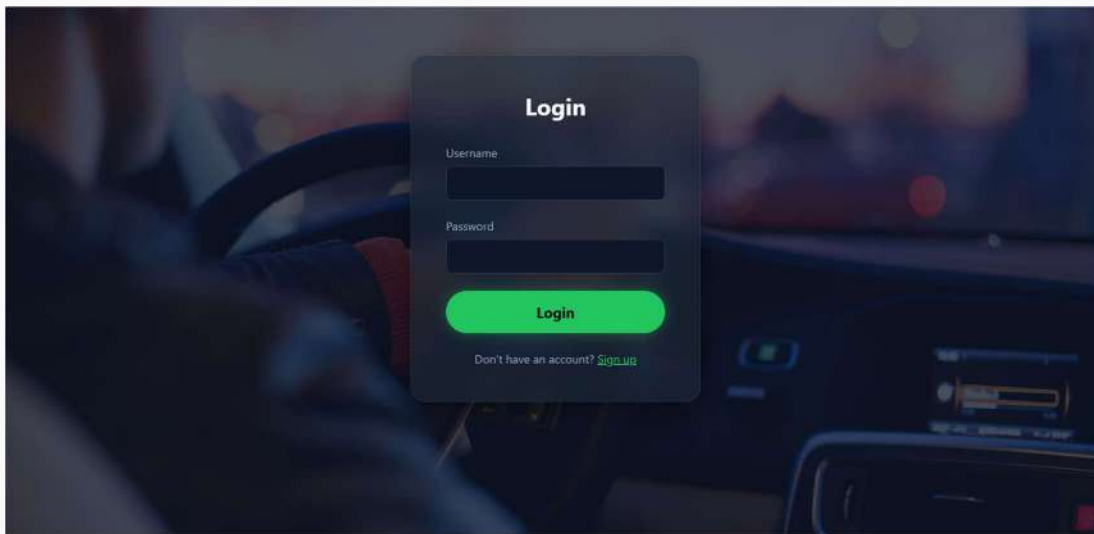
- ▶ A modern web browser (Chrome, Firefox, Edge, or Safari)
- ▶ An active internet connection
- ▶ A registered account on the system

2. Getting Started

2.1 Accessing the Application

Open your web browser and navigate to the application URL provided by your system administrator.

SCREENSHOT

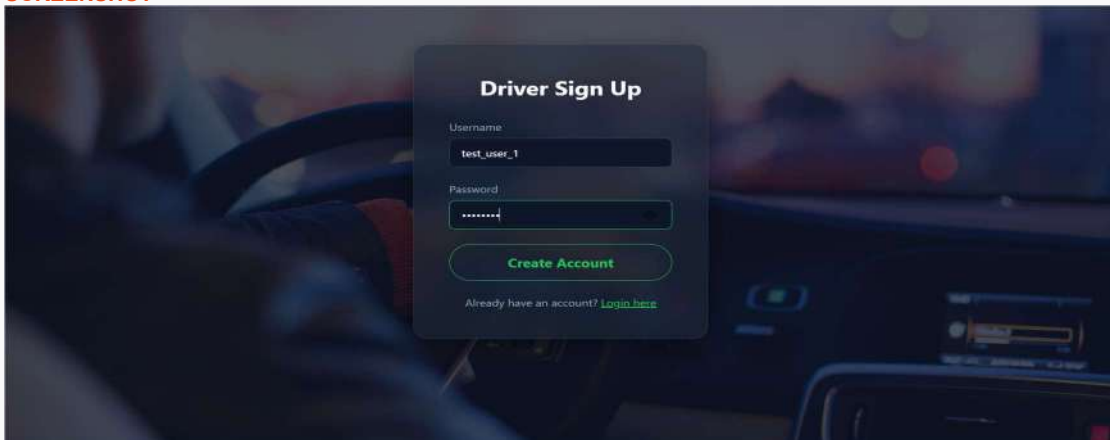


2.2 User Registration

New users must register before accessing the system:

- ▶ Click the **Create Account** button on the login page.
- ▶ Fill in your username and password.
- ▶ Submit the form and you are ready to sign in

SCREENSHOT

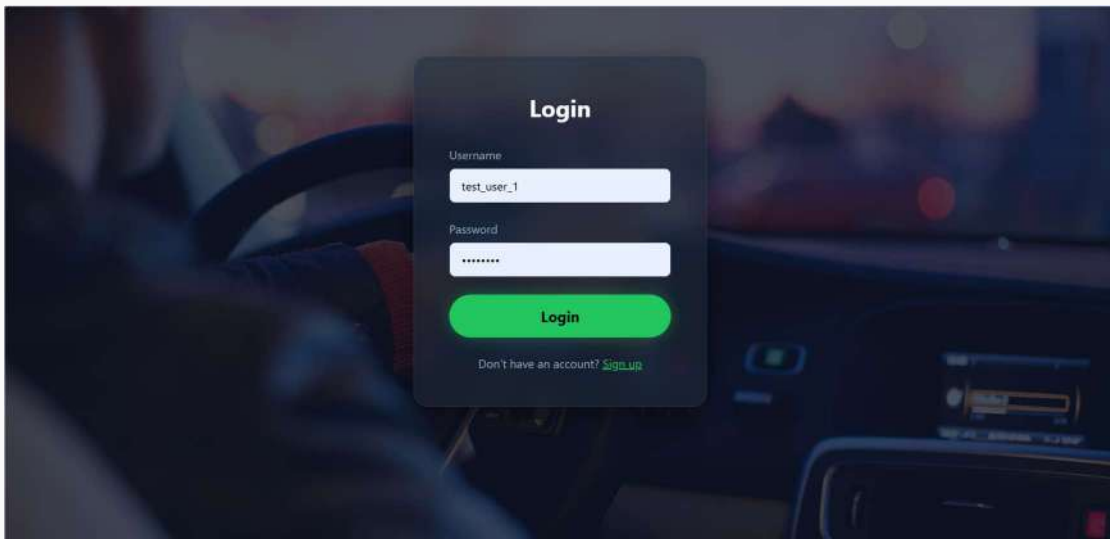


NOTE *Username must be unique. Contact your administrator if you face issues during registration.*

2.3 Logging In

- ▶ Enter your username and password on the login page.
- ▶ Click **Login**.
- ▶ You will be redirected to your dashboard.

SCREENSHOT



2.4 Logging Out

Click the **Logout** button in the top navigation bar. Your session will be terminated and you will be returned to the login page.

SCREENSHOT

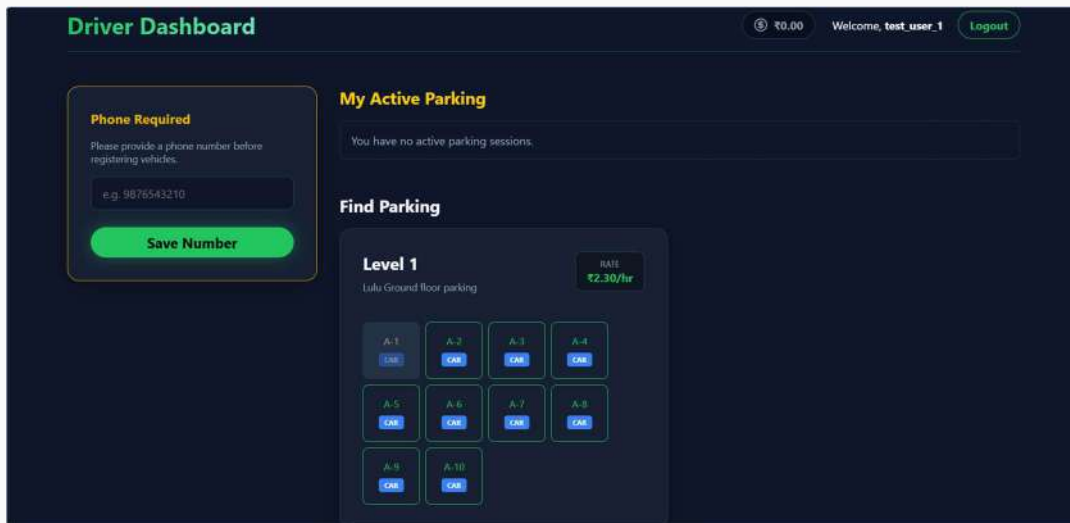


3. Driver Guide

3.1 Driver Dashboard

After login, Drivers land on their dashboard showing active sessions, due fees, and registered vehicles at a glance.

SCREENSHOT



3.2 Managing Vehicles

3.2.1 Adding a Vehicle

- ▶ Navigate to **My Vehicles** from the dashboard.
- ▶ Register user's phone number.
- ▶ Click **Add Vehicle**.
- ▶ Enter the License Plate, Vehicle Type, and Make/Model.
- ▶ Click **Save**.

SCREENSHOT

My Vehicles

No vehicles registered.

Add New Vehicle

License Plate

KL 23 J 2134

Vehicle Type

Car

Make/Model

Creta

Add Vehicle

3.2.2 Viewing Registered Vehicles

All registered vehicles are listed under **My Vehicles** with their license plate, type, and make/model.

SCREENSHOT

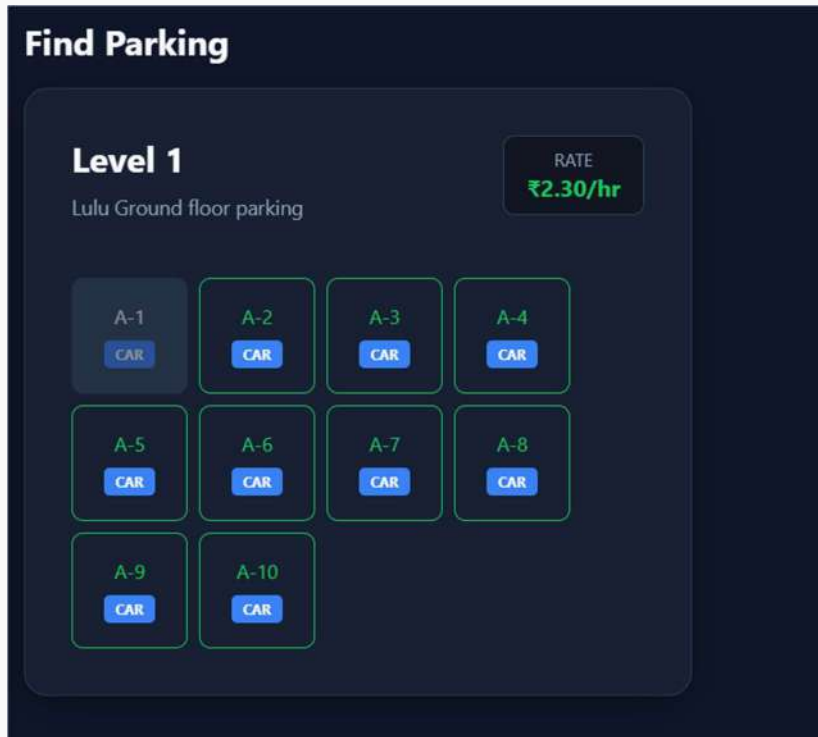
My Vehicles

KL 23 J 2134 Creta	CAR	Delete
KL 34 E 7896 Toyota Corrolla	CAR	Delete

3.3 Browsing Parking Lots

- ▶ Click **Find Parking** from the navigation menu.
- ▶ All lots are listed with name, address, total slots, and price per hour.
- ▶ Lots with no available slots are clearly indicated.

SCREENSHOT



3.4 Reserving a Parking Slot

- ▶ Click on a parking lot to view its slots.
- ▶ Select an available slot (status: Available).
- ▶ Choose your registered vehicle from the dropdown.
- ▶ Click **Reserve Slot** to confirm.

SCREENSHOT

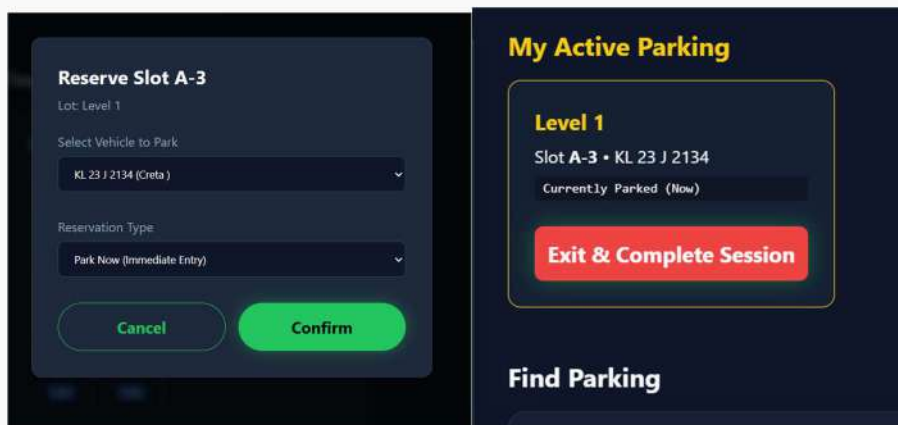


NOTE Slot availability updates in real time. Atomic transactions prevent any double-booking.

3.5 Viewing Active Session

Your active session is displayed on the dashboard, showing the lot name, slot number, entry time, and vehicle in use.

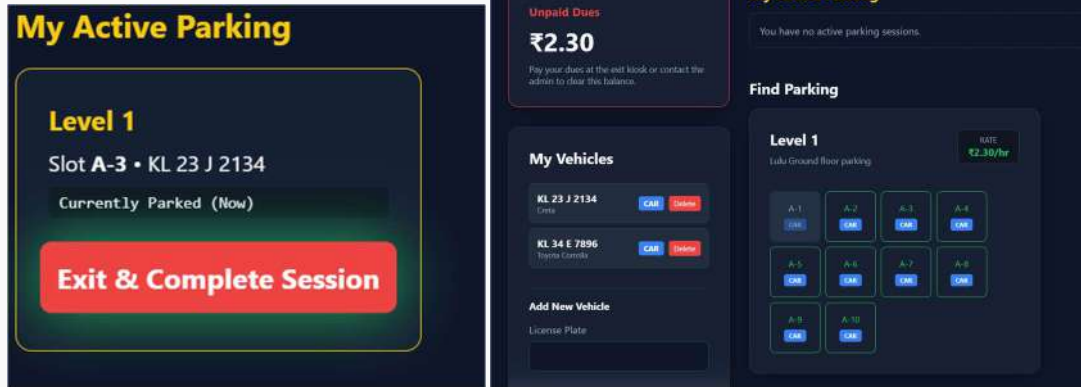
SCREENSHOT



3.6 Releasing a Slot

- ▶ Go to your active session on the dashboard.
- ▶ Click **Release Slot**.
- ▶ The system calculates duration and fees, then updates your due fees balance.

SCREENSHOT



3.7 Viewing Due Fees

Your outstanding balance (due_fees) is shown on your dashboard, updated automatically after each session ends.

SCREENSHOT

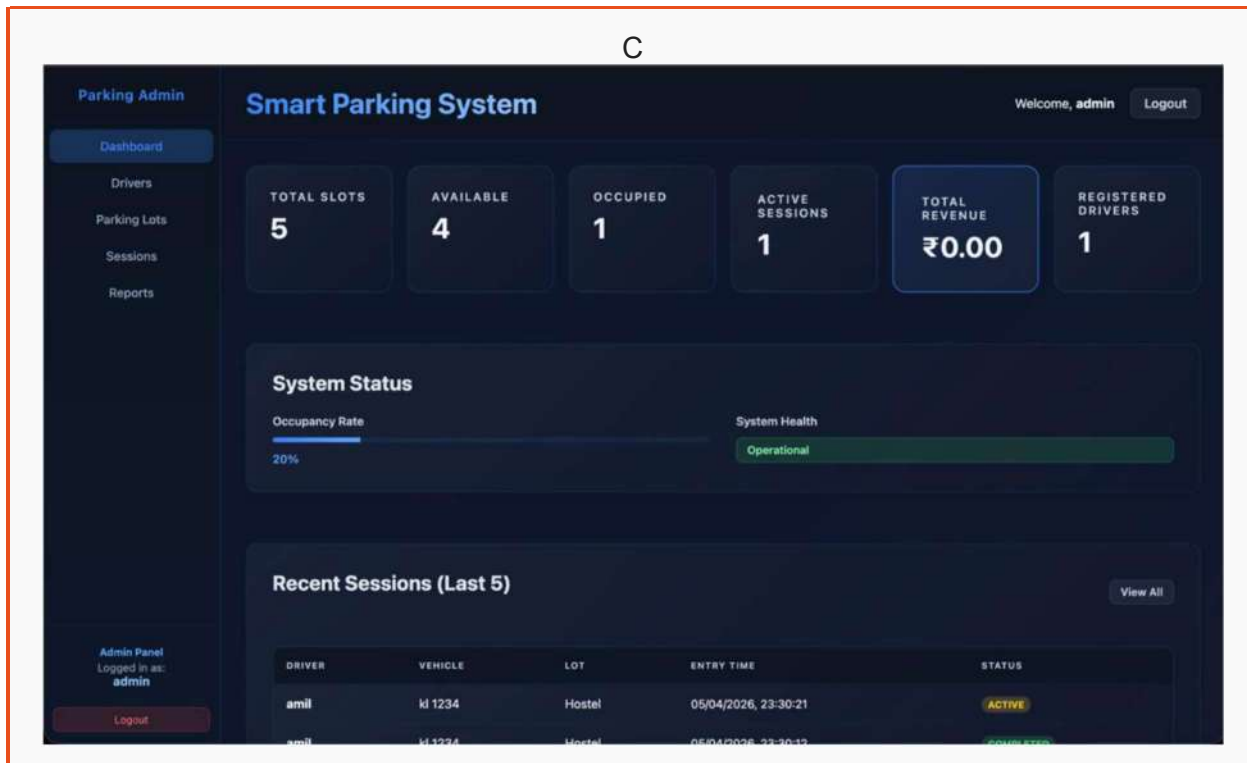


3.8 Session History

Past sessions are listed under **Session History**, showing entry/exit times, duration, fee charged, lot, and slot.

4.1 Admin Dashboard

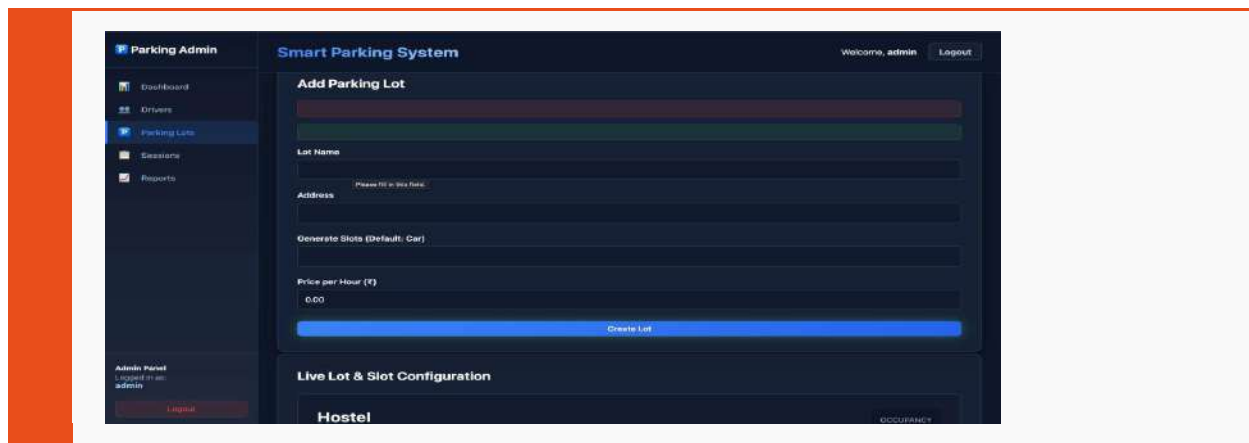
The Admin Dashboard summarises the entire system — total parking lots, active sessions, registered users, and total outstanding dues.



4.2 Managing Parking Lots

4.2.1 Adding a Parking Lot

- ▶ Navigate to Parking Lots from the admin menu.
- ▶ Enter the Lot Name, Address, Total Slots, and Price Per Hour.
- ▶ Click **Create Lot** — slots are auto-generated by the system.



4.2.2 Editing a Parking Lot

Click **Edit** next to a lot, update the required fields, and click **Save**.

4.2.3 Deleting a Parking Lot

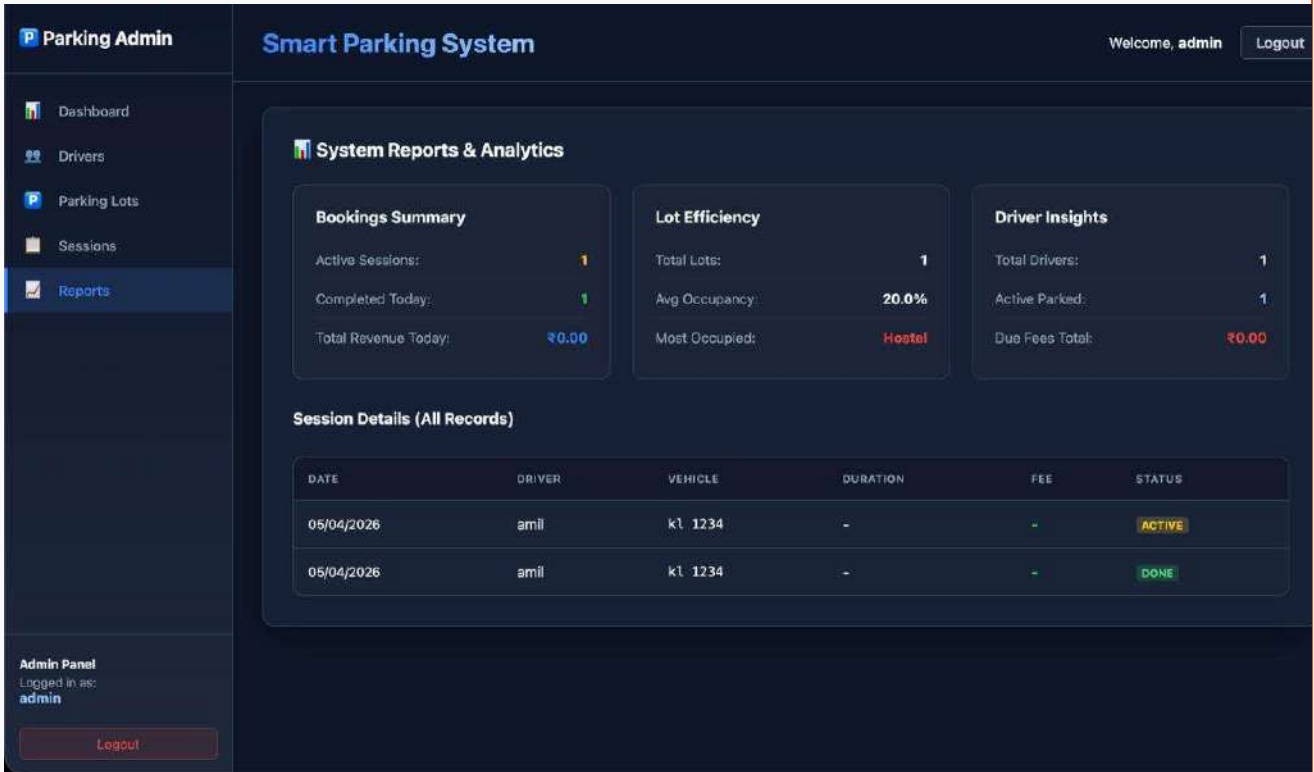
Click **Delete** on a parking lot to remove it along with all its associated slots. This action is irreversible.

NOTE Deleting a lot cascade-deletes all its slots. Ensure no active sessions exist in the lot before deletion.

4.3 Managing Parking Slots

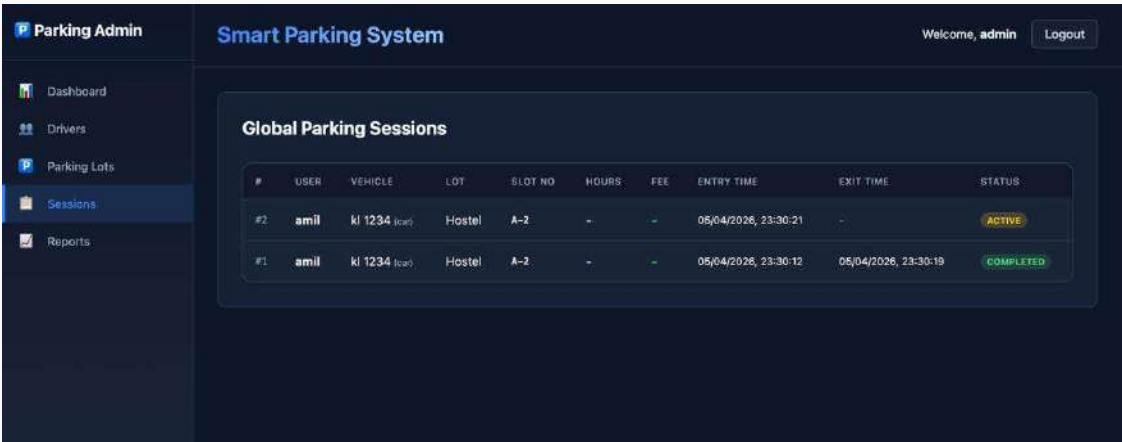
Admins can view all slots within a lot, including current status (Available, Occupied, or Reserved) and any assigned driver/vehicle.

SCREENSHOT



4.4 Monitoring Active Sessions

The Active Sessions panel shows all ongoing sessions across all lots — driver, vehicle, lot, slot, entry time, and running duration.



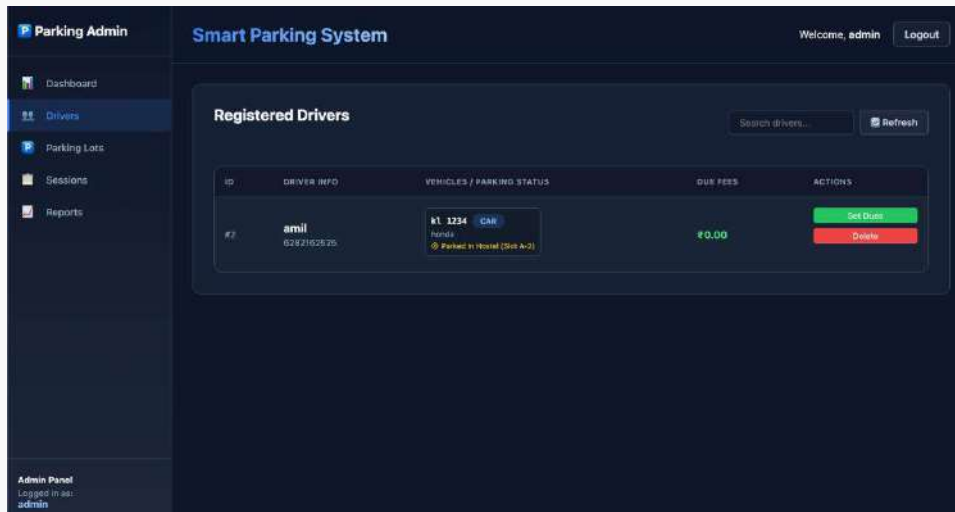
4.5 Managing Users

The user list shows username, role, phone number, and due fees balance for every registered user.

4.6 Manual Fee Override

- ▶ Navigate to the **User list** and locate the driver.
- ▶ Click **Adjust Fees**.
- ▶ Enter the corrected fee amount and a reason.
- ▶ Click **Save** — the balance updates immediately.

SCREENSHOT



NOTE This feature is for resolving billing discrepancies or for fining users in the case of any damages or inconveniences caused by them. All adjustments are logged.

5. Billing and Fees

5.1 Fee Calculation

Fees are computed automatically when a driver releases a slot:

⚡ Formula

Fee Charged = Duration (hours) × Price Per Hour (of the parking lot)

The fee is added to the driver's due_fees balance.

The fee is permanently stamped onto the session record (fee_charged).

5.2 Due Fees Balance

A driver's outstanding balance lives in the `due_fees` field on their account. It increases when sessions end and can only be reduced by an Admin override.

6. Troubleshooting

Issue	Solution
Cannot log in	Check your username and password. Contact admin if your account is locked.
Slot shows occupied but is empty	Ask your admin to manually reset the slot status.
Fees seem incorrect	Request an admin to review and apply a manual fee override.
Cannot add a vehicle	Ensure all fields (license plate, type, model) are filled in.
Page not loading	Check your internet connection and refresh the browser.
Session not ending after releasing slot	Refresh and check session history. Contact admin if issue persists.

7. Glossary

Admin — A user role with full system access — lot management, user management, and billing overrides.

Driver — A user role that registers vehicles, finds parking, and manages sessions.

Parking Lot — A physical parking location with multiple slots, a price per hour, and an address.

Parking Slot — An individual space within a lot. Status: Available, Occupied, or Reserved.

Parking Session — A record of a driver occupying a slot, from entry to exit, including fee charged.

Due Fees — The cumulative unpaid parking balance on a driver's account.

RBAC — Role-Based Access Control — restricts actions based on the user's role.

JWT — JSON Web Token — a secure token used to authenticate user sessions.

ACID — Atomicity, Consistency, Isolation, Durability — database transaction guarantees.

API — Application Programming Interface — the backend layer between frontend and database.
